

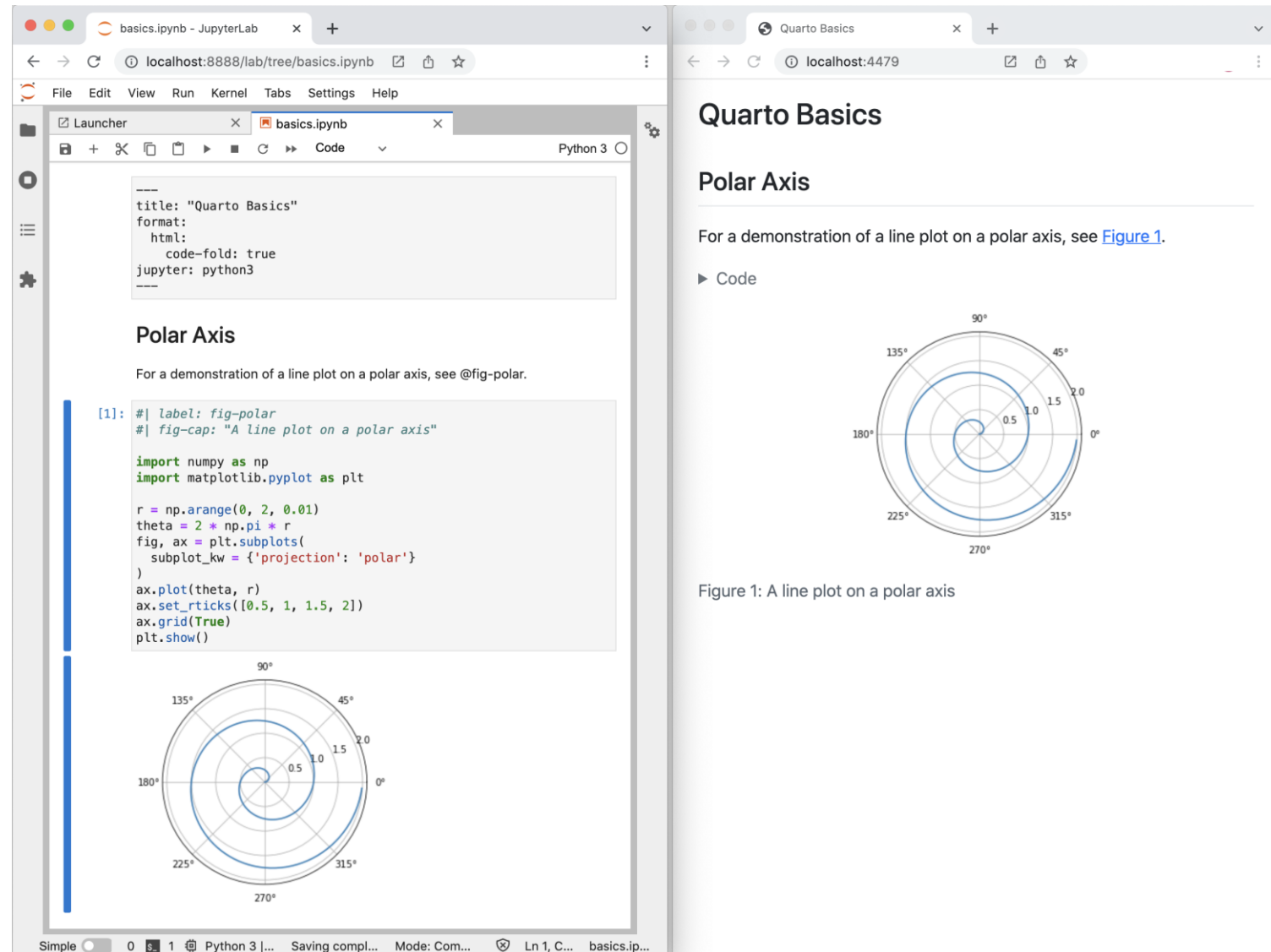


Jupyter Notebooks & Google Colab

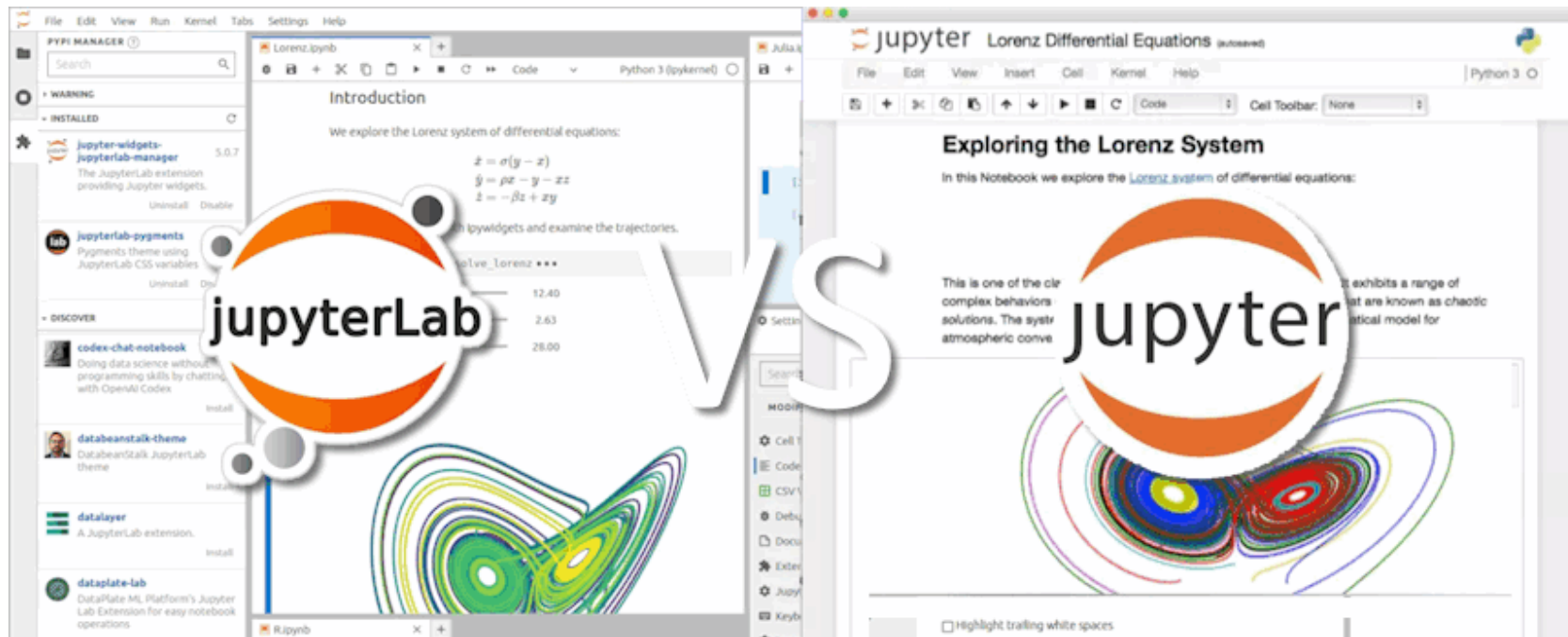
- Jupyter Notebooks : what they are
- Jupyter Notebooks hosted e local: GColab
- GColab: Runtime types
- GColab: Connecting to GDrive
- GColab: Creating a notepad
- GColab: insert and execute code blocks
- GColab: Variable control

Jupyter Notebooks: What They Are

- Jupyter Notebooks are a versatile data science tool used for scientific purposes, such as data analysis, data preparation and ML model training
- They are web based, that is, they can be used via a web browser
- They mix blocks of text in which to write code to blocks of text in which to write markdown notes



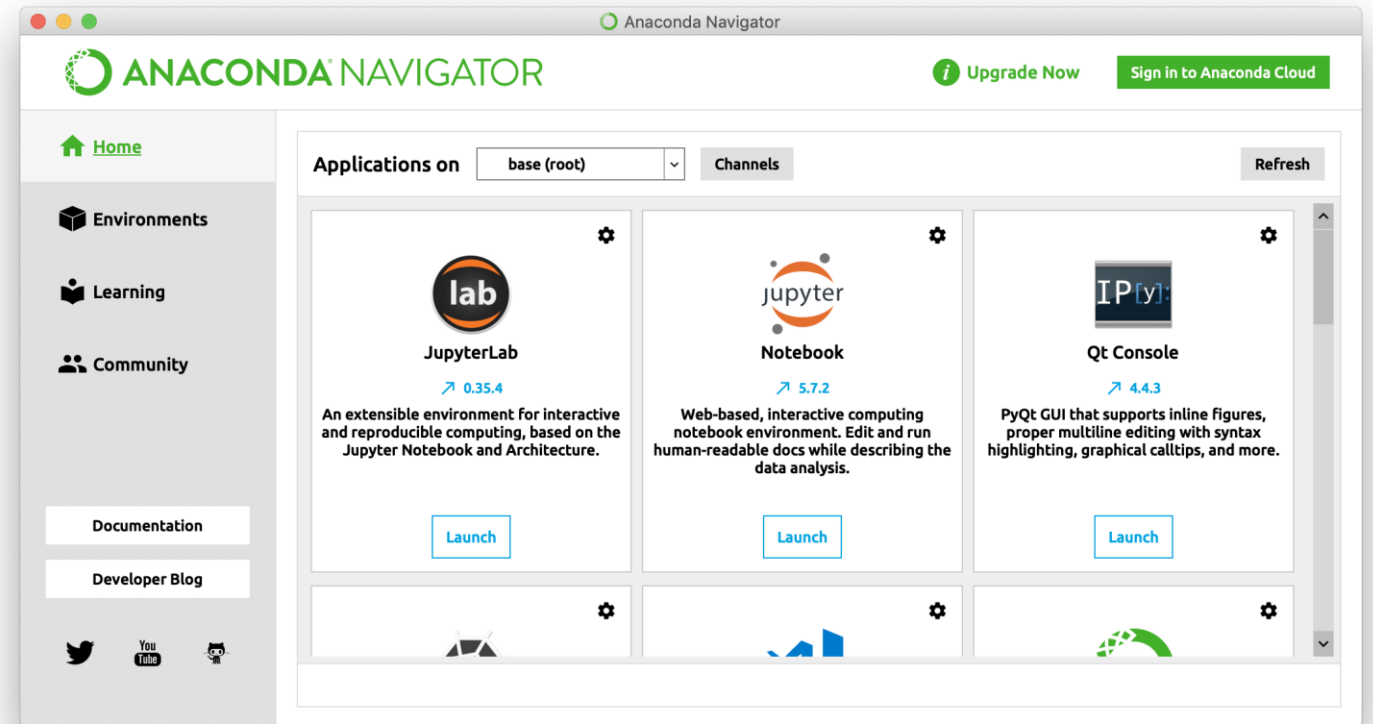
Jupyter Notebooks vs Jupyter Lab



- An improved version of the **Jupyter Notebook** is **Jupyter Lab**.
- Jupyter Notebooks and Jupyter Lab are also distributed through other packages, such as Anaconda.
- The Jupyter Notebooks can be used in two ways: *hosted* and *local*.
- The hosted versions are *offered by third parties* and freely accessible. An example is **Google Colab**.

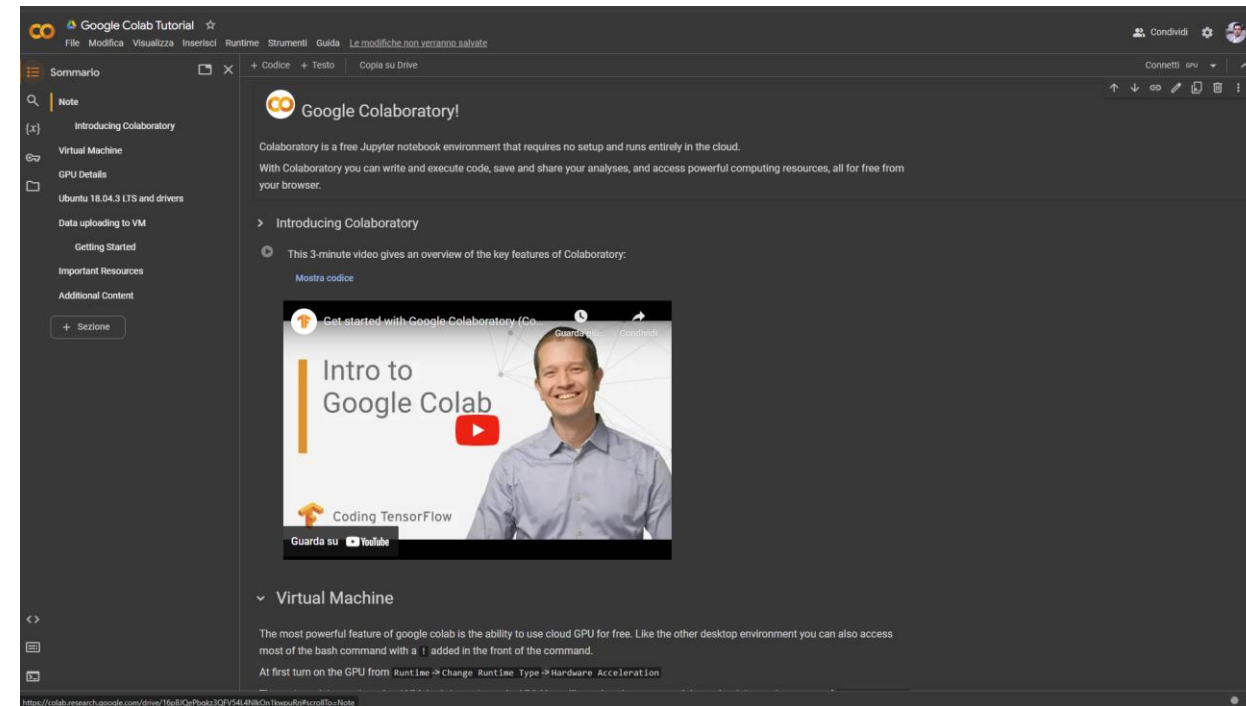
Jupyter Lab: Anaconda

- 1.Environment management:** Anaconda allows you to create isolated Python environments for separate projects.
- 2.Kernel:** In Jupyter Lab, each notebook is associated with a Python kernel that runs the code.
- 3.Package management:** Allows for simple installation and management of Python packages.
- 4.Jupyter Lab Integration :** Anaconda integrates with Jupyter Lab for data exploration and interactive development.
- 5.Results display:** Jupyter Lab displays output as charts and tables directly in the notebook.



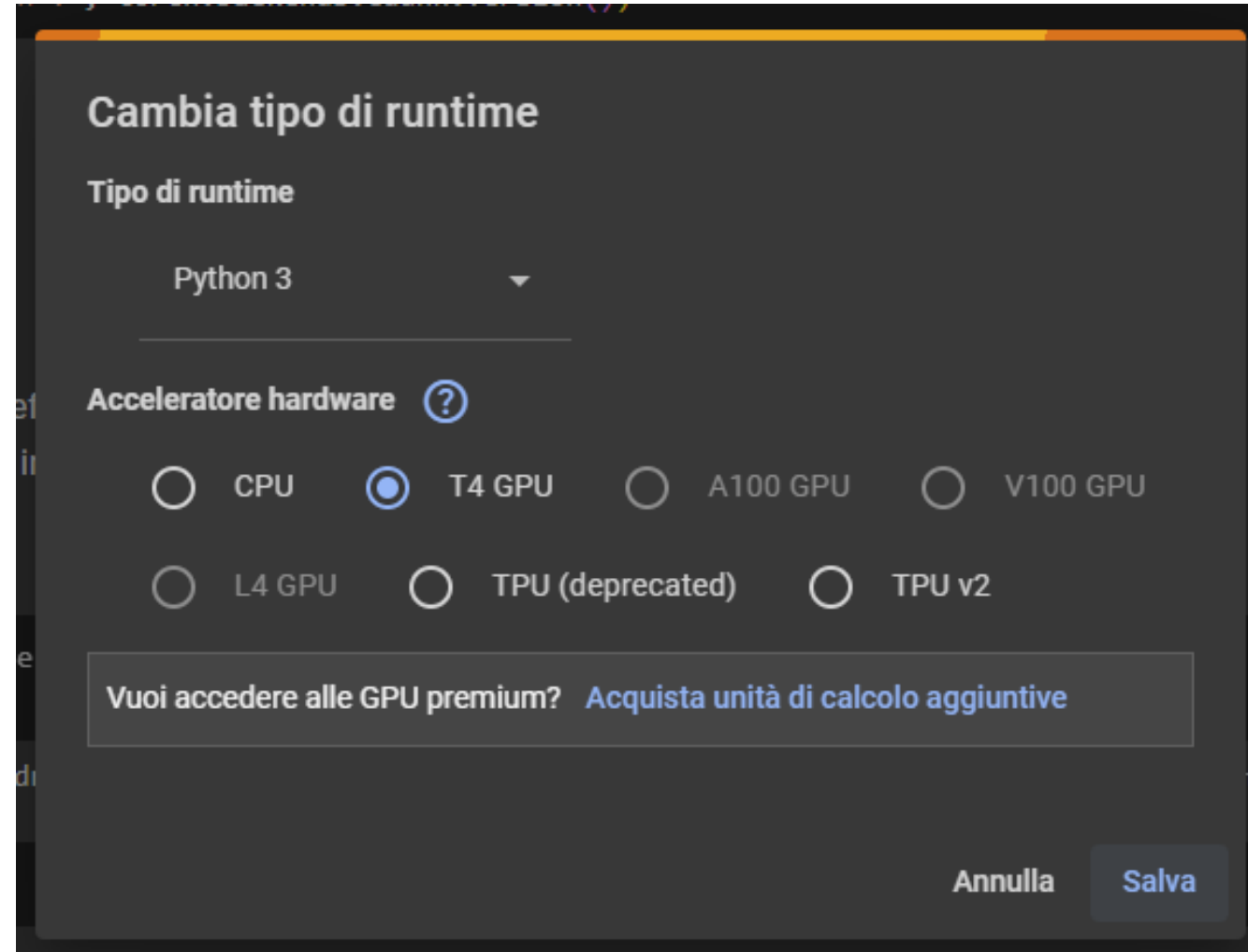
Jupyter Notebooks: Google Colab

- 1.Notebook cloud environment:** Free access to a cloud-based Jupyter environment .
- 2.GPU and TPU available:** Possibility to use powerful computing resources such as GPU and TPU for free.
- 3.Google Drive integration:** Easy to save and upload notebooks and files to Google Drive.
- 4.Pre-installed libraries:** Inclusion of popular libraries for machine learning and data science.
- 5.Real-time collaboration:** Support for simultaneous collaboration on notebooks.
- 6.BigQuery Data Access:** Integration to query large datasets.
- 7.Extensions and integration with GitHub:** Possibility to extend the functionality and integrate with GitHub for project management.



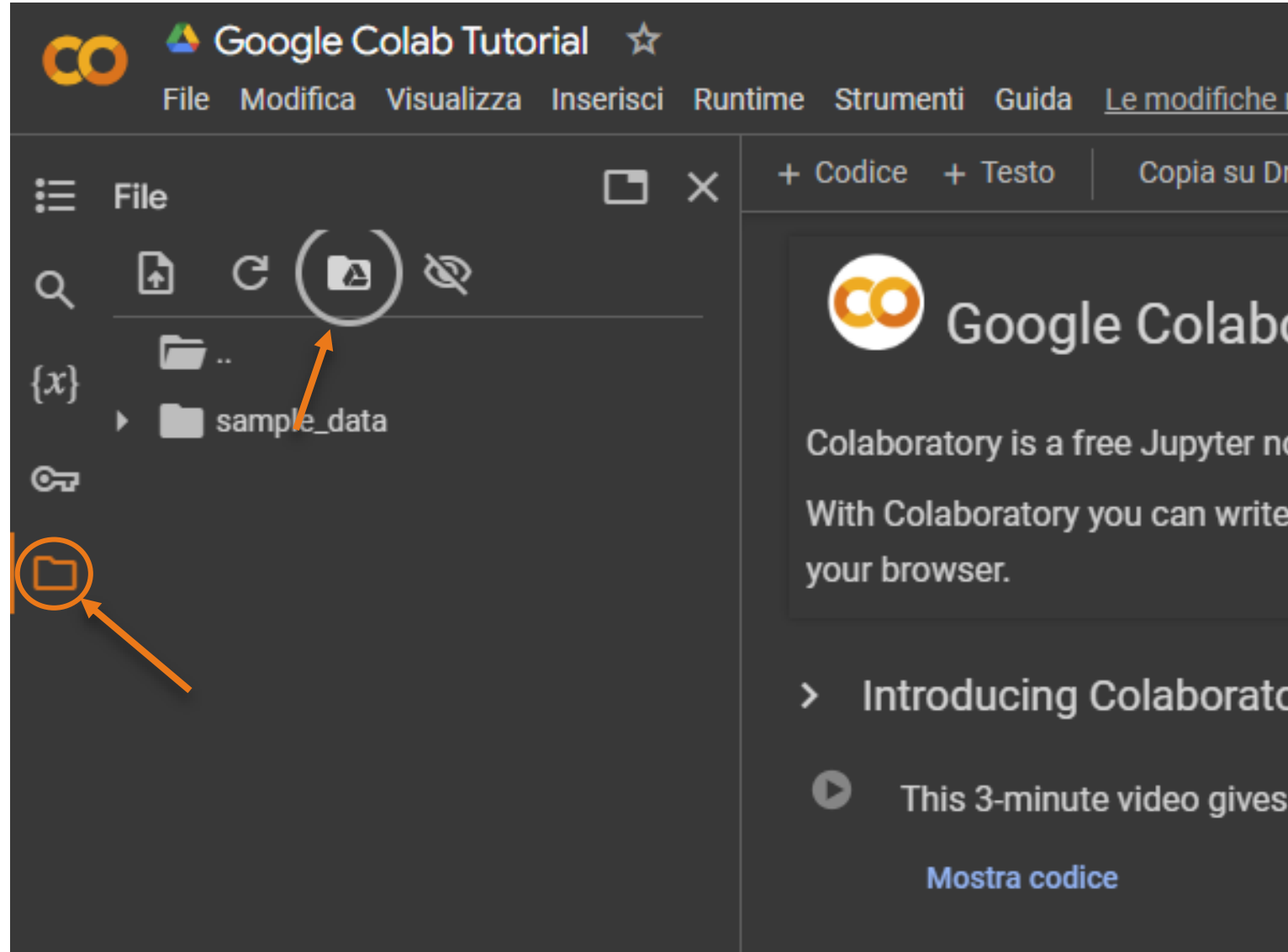
GColab: Runtime types

- GColab offers a virtual machine in the cloud fully accessible from the notebook
- The machine is connectable to your Google account via GDrive
- The machine comes with several types of runtime, including some **specialized for training ML models**, such as GPU or TPU.



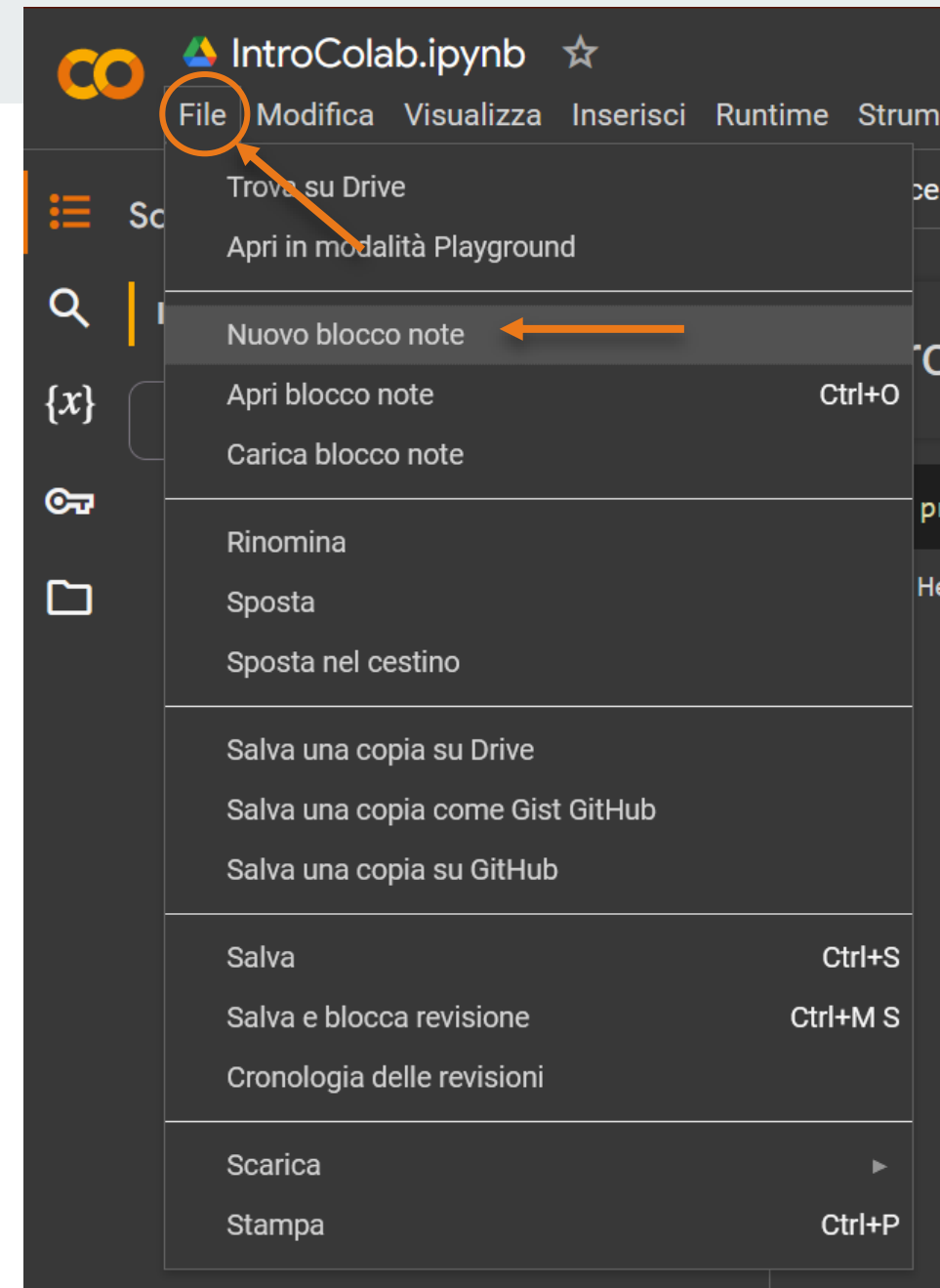
GColab: Connection to GDrive

- To connect the environment with **GDrive**, click on the folder icon on the left
- Then click the folder symbol with GDrive follow the instructions

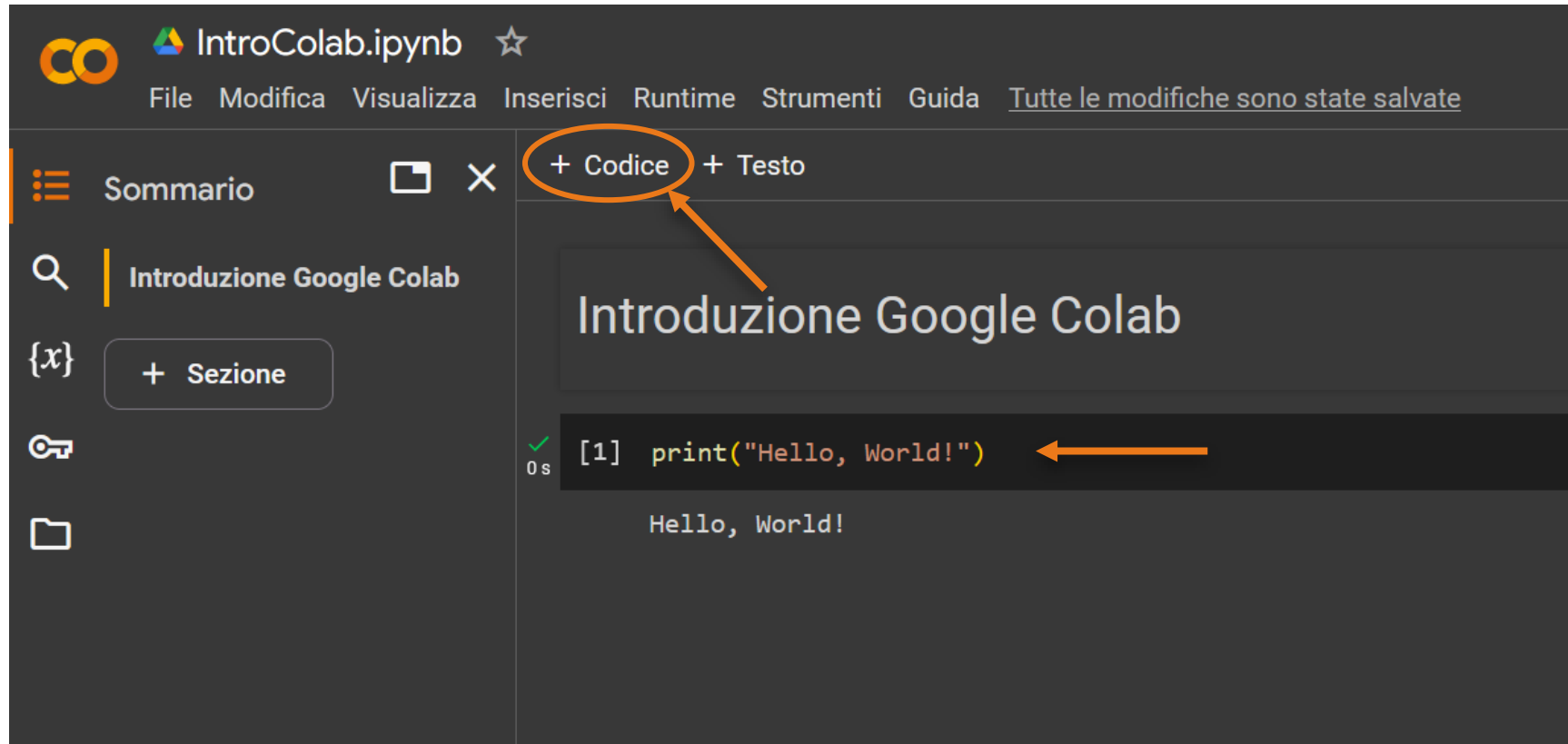


GColab: Creating a notepad

- To split the work into different files you must click «File» and then New notepad to create a file `.ipynb`
- Once created, you can work on this file by inserting text and Python code to run.
- To upload a file already present in our Drive you can click on «Open notebook».



GColab: Insert Code Blocks



- To write the code to run in Python click on +Code
- This command creates a text box like the one shown in the image, where you can write the code that will be executed

GColab: Execute Code Blocks

- To execute the blocks of code, click on the symbol of Play next to these
- Important note that you must execute the code blocks in the correct order so that they work exactly
- Once blocks 1 and 2 are executed, the variables are created and stored
- You can run all blocks of code in order by clicking on **Runtime -> Run all**

1

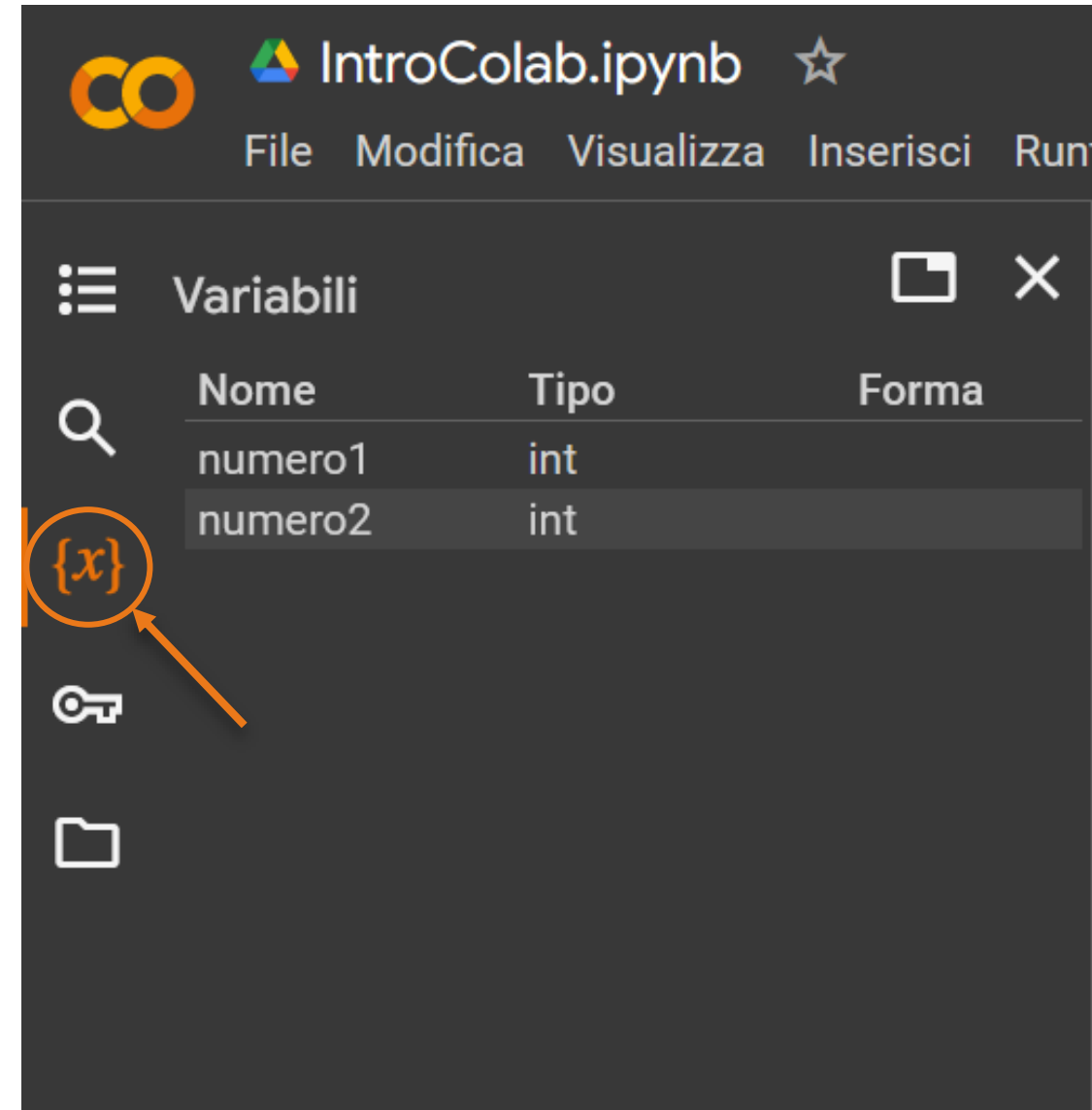
```
0s [2] print("Hello, World!")  
Hello, World!  
Creazione di una variabile  
0s [4] numero1 = 1  
Creazione di un'altra variabile  
0s [5] numero2 = 2  
Utilizzo di entrambe le variabili  
0s print(numero1 + numero2)  
3
```

2

3

GColab: Variable Inspector

- You can check the value of the variables created by clicking on the highlighted symbol
- To view the value, click on the variable concerned



GColab: from Design to Production



- GColab and JupyterNB are convenient to describe and experiment on data.
- Tools are also widely used to define the necessary data pipelines at the design stage, from their receipt to model training.
- Once pipelines are defined and tested, they are typically moved to Production and enter the DevOps process.